

## TD 1

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**Exercice 1 – Bayesian Classifier (Author : F. Rossi)**

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**Q 1.1** We consider the dataset of votes cast by members of the United States House of Representatives in 1984 on 16 major proposals. Each individual is a member of the House described by 17 nominal variables. The variable *Party* takes the modalities Democrat and Republican. The other variables,  $V_1$  to  $V_{16}$ , represent the votes and take the values YES, NO, and ABSTAIN (for a missed vote). There are 267 Democratic representatives and 168 Republican representatives.

|             |          | NO  | ABSTAIN | YES |           |          |     |    |     |
|-------------|----------|-----|---------|-----|-----------|----------|-----|----|-----|
| Republicans | $V_1$    | 134 | 3       | 31  | Democrats | $V_1$    | 102 | 9  | 156 |
|             | $V_2$    | 73  | 20      | 75  |           | $V_2$    | 119 | 28 | 120 |
|             | $V_3$    | 142 | 4       | 22  |           | $V_3$    | 29  | 7  | 231 |
|             | $V_4$    | 2   | 3       | 163 |           | $V_4$    | 245 | 8  | 14  |
|             | $V_5$    | 8   | 3       | 157 |           | $V_5$    | 200 | 12 | 55  |
|             | $V_6$    | 17  | 2       | 149 |           | $V_6$    | 135 | 9  | 123 |
|             | $V_7$    | 123 | 6       | 39  |           | $V_7$    | 59  | 8  | 200 |
|             | $V_8$    | 133 | 11      | 24  |           | $V_8$    | 45  | 4  | 218 |
|             | $V_9$    | 146 | 3       | 19  |           | $V_9$    | 60  | 19 | 188 |
|             | $V_{10}$ | 73  | 3       | 92  |           | $V_{10}$ | 139 | 4  | 124 |
|             | $V_{11}$ | 138 | 9       | 21  |           | $V_{11}$ | 126 | 12 | 129 |
|             | $V_{12}$ | 20  | 13      | 135 |           | $V_{12}$ | 213 | 18 | 36  |
|             | $V_{13}$ | 22  | 10      | 136 |           | $V_{13}$ | 179 | 15 | 73  |
|             | $V_{14}$ | 3   | 7       | 158 |           | $V_{14}$ | 167 | 10 | 90  |
|             | $V_{15}$ | 142 | 12      | 14  |           | $V_{15}$ | 91  | 16 | 160 |
|             | $V_{16}$ | 50  | 22      | 96  |           | $V_{16}$ | 12  | 82 | 173 |

**Q 1.1.1** How many different values are possible for the vote vector?

**Q 1.1.2** Given the vote vector of a representative :

$V = (\text{YES}, \text{NO}, \text{ABSTAIN}, \text{YES}, \text{NO}, \text{YES}, \text{YES}, \text{YES}, \text{NO}, \text{NO}, \text{YES}, \text{NO}, \text{NO}, \text{NO}, \text{NO}, \text{YES})$

How can we estimate whether they are Republican or Democrat?

**Q 1.2** We consider two populations, men  $H$  with an average height of 1.74m and a standard deviation of 0.07m, and women  $F$  with an average height of 1.62m and a standard deviation of 0.065m (data from INSEE 2001). The population  $H$  contains  $|h|$  individuals, and the population  $F$  contains  $|f|$  individuals. We assume that the height distributions are Gaussian within each subpopulation.

A person is chosen randomly and uniformly from the total population. We want to determine their subpopulation based solely on their height : this is a classification problem based on a continuous variable.

**Q 1.2.1** Define the random variable  $G$  representing the gender of a randomly chosen person. Provide the distribution of  $G$ .

**Q 1.2.2** Define the random variable  $T$  representing the height of a randomly chosen person. Provide the density of  $T$  and  $P(G = f|T = t)$ .

**Q 1.2.3** Provide the optimal Bayesian classifier.

**Q 1.2.4** Assume  $|h| = |f|$ . Specify the decisions made by the optimal classifier and interpret this decision strategy.

**Q 1.3** How many parameters does the Bayesian classifier have?

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**Exercice 2 – Bayesian Classifier**

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Let  $\mathcal{X}$  be a feature space in  $\mathbb{R}^d$  and  $\mathcal{Y}$  the set of labels  $\{y_1, \dots, y_l\}$ .

**Q 2.1** Recall what a Bayesian classifier is.

**Q 2.2** Express the error made by the Bayesian classifier at a point  $\mathbf{x}$ . Is this error minimal?

**Q 2.3** Let  $\lambda(y_j, y_i)$  be the cost of predicting label  $y_j$  instead of  $y_i$ . What are the values of  $\lambda$  in the case of 0-1 loss? Provide examples of asymmetric costs and their use cases.

**Q 2.4** What is the expression of the risk  $R(y_i|\mathbf{x})$  of predicting  $y_i$  given  $\mathbf{x}$  as a function of  $\lambda$  and posterior probabilities? What about the 0-1 loss case?

**Q 2.5** Provide the expression for the risk  $R(f)$  of a classifier  $f$  over  $\mathcal{X}$ .

**Q 2.6** In the binary case, express the decision criterion in terms of  $\lambda$ , posterior probabilities, class distributions, and likelihoods.

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### Exercise 3 – Density Estimation

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**Q 3.1** Give the density estimate  $p_{\mathcal{B}}$  of a random variable  $X$  inside a region of interest  $\mathcal{B}$  of volume  $V$ , based on a number  $k$  of observed samples in this region out of  $n$  drawn samples.

**Q 3.2** Let  $X$  be a random variable in  $\mathcal{X}$ . We want to estimate the density  $p_X$  of this variable from a set of observations  $\mathcal{X}_o$ . Describe how to proceed to make this estimation using the histogram method.

**Q 3.3** Discuss kernel density estimation methods.

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### Exercise 4 (4 points) – Risk

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Let the result of a medical test be expressed as a real number  $x \in \mathbb{R}$ , and let there be two classes,  $y_-$  and  $y_+$ , representing "not sick" and "sick" respectively. We know that :

$$P(y = y_+|x) = \begin{cases} 0 & \text{if } x \leq 0, \\ x & \text{if } 0 \leq x \leq 1, \\ 1 & \text{if } x \geq 1. \end{cases}$$

We use a classifier of the form :

$$f_{\theta}(x) = \begin{cases} y_+ & \text{if } x > \theta, \\ y_- & \text{if } x \leq \theta. \end{cases}$$

**Q 4.1** Give the expression for the 0-1 loss and the risk at a point  $x^0$  as a function of  $\theta$ .

**Q 4.2** Suppose that the result of the test  $x$  is uniformly distributed in  $[-1, 1]$ . What is the optimal classifier? What is the minimum risk value?

**Q 4.3** It is known that it is three times more costly to classify a sick person as not sick than the reverse. Give the associated cost function, the new risk formulation, and the optimal classifier.

**Q 4.4** Could we do better with a Bayesian classifier? What about a naive Bayesian classifier?